

Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

Ronald Melvin R. Rosas Marcus Rafael Tiongson

AI 201 Project Defense

The Problem

Fuel estimation in Philippine agricultural mechanization relies on static coefficients.

- ▶ Generalized coefficients ignore machine variability
- ▶ No localized field, soil, or operator effects
- ▶ Nonlinear interactions invisible to deterministic formulas
- ▶ Result: weak planning, missed efficiency gains

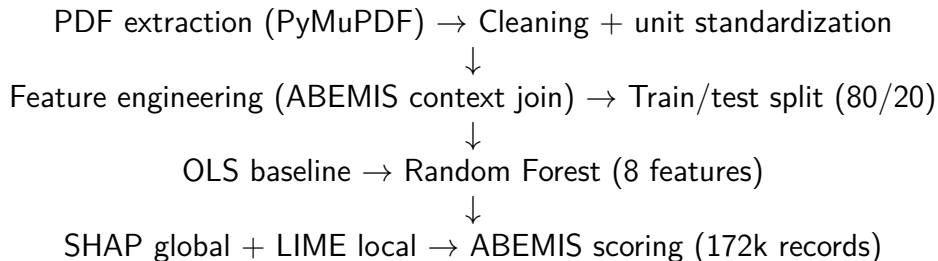
Goal

- ▶ Develop an ML model for fuel-requirement estimation
- ▶ Compare against OLS baseline using R^2 , MAE, RMSE
- ▶ Score national ABEMIS deployments for regional fuel planning
- ▶ Interpret with SHAP (global) and LIME (local)

The Data

- ▶ AMTEC test reports → **378** cleaned training records
- ▶ ABEMIS inventory → **246,741** fuel-relevant records
- ▶ Coverage: **all 17** Philippine regions
- ▶ **8** RF features: `power_kw`, `year`, three ABEMIS context features, three categorical-encoded (machinery type, family, subset)
- ▶ Target: fuel consumption (L/hr)

Pipeline



Methodology Highlights

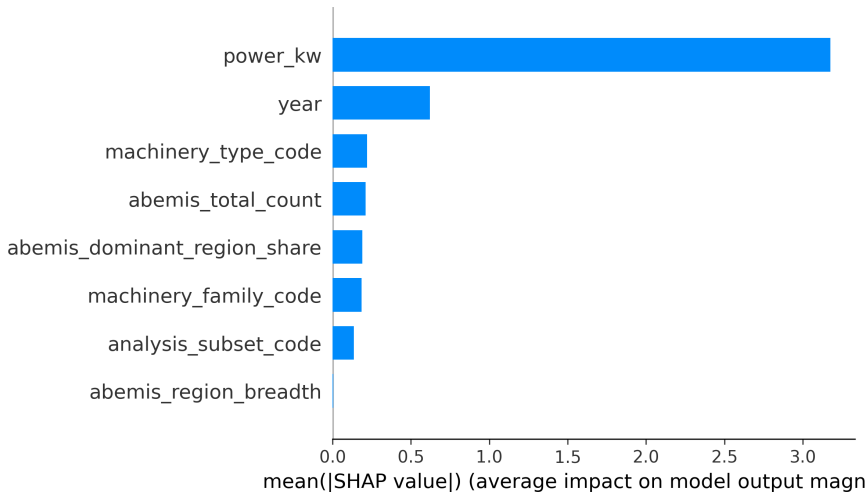
- ▶ **OLS baseline** — linear/quadratic; per-type fallback hierarchy (type → family → global)
- ▶ **Random Forest** — scikit-learn defaults, 80/20 split, 8 features, 5-fold CV
- ▶ **Interpretability** — SHAP TreeExplainer (bar + beeswarm), LIME tabular explainer (10 example records)

Results — Model Comparison ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.761	1.683	2.640
OLS Hierarchical Fallback	0.722	1.872	2.847
Random Forest	0.911	1.090	1.611

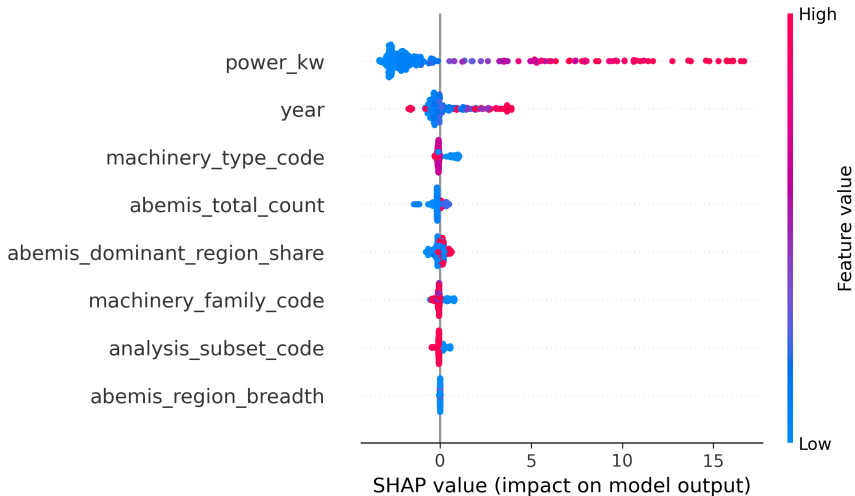
Random Forest captures non-linear power-fuel interactions OLS cannot.

SHAP — Global Feature Importance



power_kw dominates at ~76% importance.

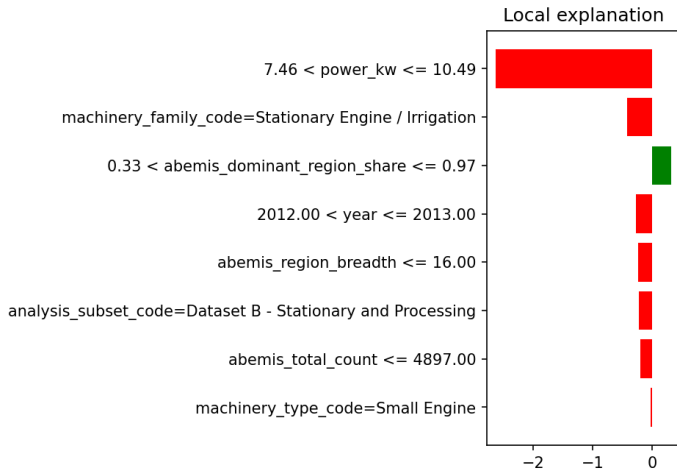
SHAP — Direction & Magnitude



Higher power → higher predicted fuel; context features add nuance.

LIME — A Local Explanation

LIME - Sample 374 | Small Engine | Actual=5.240 Pred=3.746



Per-record explanation builds trust for BAFE engineers.

ABEMIS Regional Scoring

- ▶ **172,265** of 246,741 records successfully scored (**70%** coverage)
- ▶ Excluded: 62,808 missing Rated Power, 11,668 unmappable type
- ▶ Outputs: per-region, per-region $\times$ type, per-record fuel estimates
- ▶ Enables data-driven mechanization planning at national scale

Limitations & Future Work

- ▶ OCR retraining on 247 scanned PDFs (Tesseract binary pending)
- ▶ RAED cropping calendar integration as the Time predictor
- ▶ Expand ABEMIS $\rightarrow$ AMTEC type mapping for $\sim 19,674$ unmapped rows
- ▶ Validation against in-field telemetry from a pilot deployment

Thank you

Ronald Melvin R. Rosas
Marcus Rafael Tiongson

Repository: `fuel-abemis`